

$$\begin{aligned}
 \underline{C1:} X(\omega) &= \cos^2 \omega = \frac{1}{2} (1 + \cos 2\omega) \\
 &= \frac{1}{2} \left[1 + \frac{1}{2} (e^{j2\omega} + e^{-j2\omega}) \right] \\
 &= \frac{1}{2} + \frac{1}{4} e^{j2\omega} + \frac{1}{4} e^{-j2\omega} \\
 &= \underbrace{\frac{1}{2}}_{X_1(\omega)} + \underbrace{\frac{1}{4} e^{j2\omega}}_{X_2(\omega)} + \underbrace{\frac{1}{4} e^{-j2\omega}}_{X_3(\omega)}
 \end{aligned}$$

Theo TC tuyến tính của DTFT:

$$\begin{aligned}
 x[n] &= \text{IDTFT}\{X(\omega)\} = \text{IDTFT}\{X_1(\omega)\} \\
 &\quad + \text{IDTFT}\{X_2(\omega)\} + \text{IDTFT}\{X_3(\omega)\} \\
 &= x_1[n] + x_2[n] + x_3[n]
 \end{aligned}$$

$$x_1[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\omega) e^{j\omega n} d\omega$$

$$f(x) = \text{sinc}(x)$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{2} e^{j\omega n} d\omega$$

$$= \frac{1}{4\pi} \cdot \frac{1}{jn} \cdot e^{j\omega n} \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi jn} \cdot (e^{j\pi n} - e^{-j\pi n})$$

$$= \frac{\sin(\pi n)}{2\pi n} = \frac{1}{2} \text{sinc}(n)$$

$$T+0: x_2[n] = \frac{1}{4} \text{sinc}(n+2)$$

$$x_3[n] = \frac{1}{4} \text{sinc}(n-2)$$

$$\rightarrow x[n] = \frac{1}{2} \text{sinc}(n) + \frac{1}{4} \text{sinc}(n+2) + \frac{1}{4} \text{sinc}(n-2)$$

$$f(x) = \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x} = \begin{cases} 1 (x=0) \\ 0 (x \in \mathbb{Z}, x \neq 0) \end{cases}$$

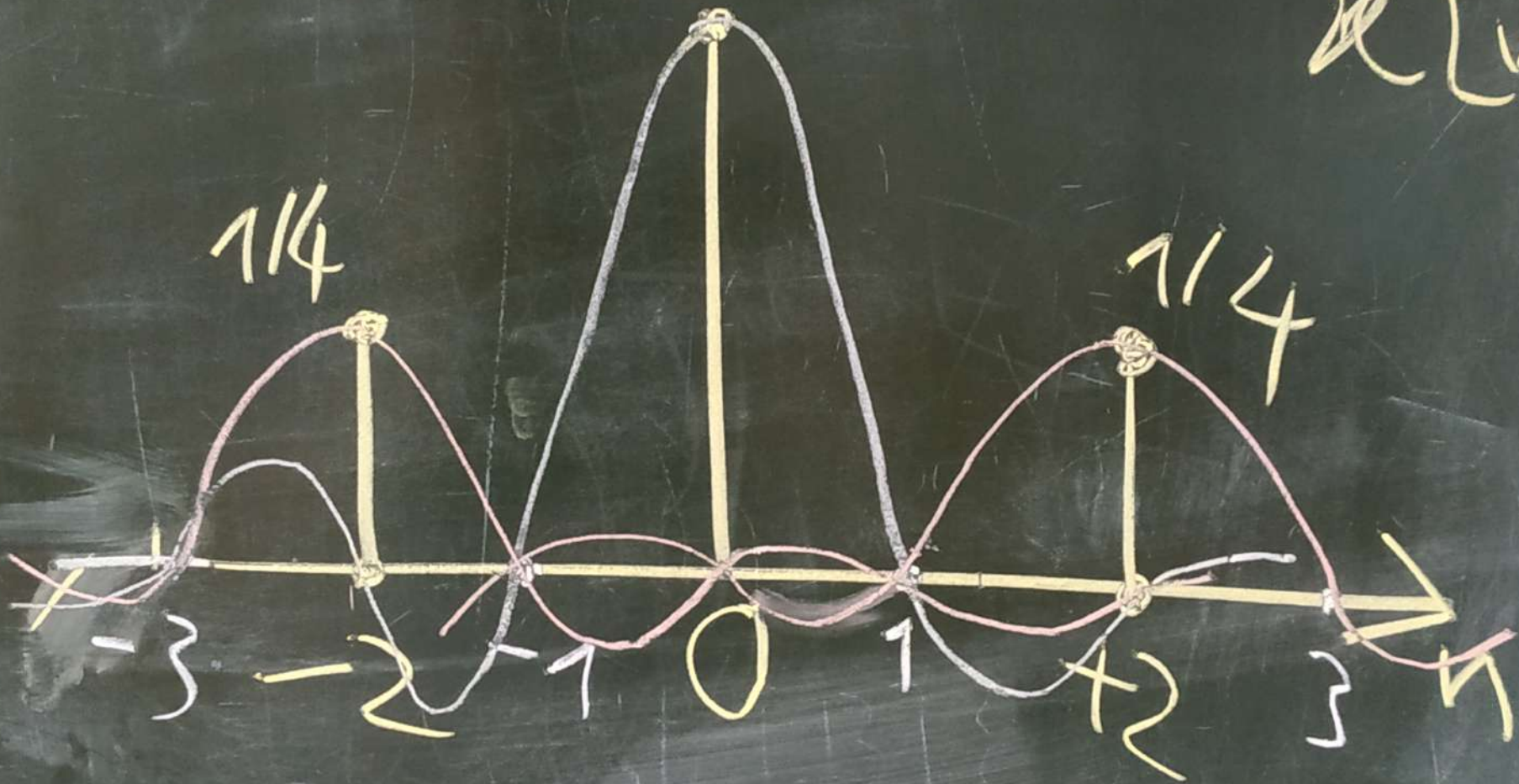
$|f(x)| \rightarrow 0$ as $x \rightarrow \pm\infty$

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$x[n]$

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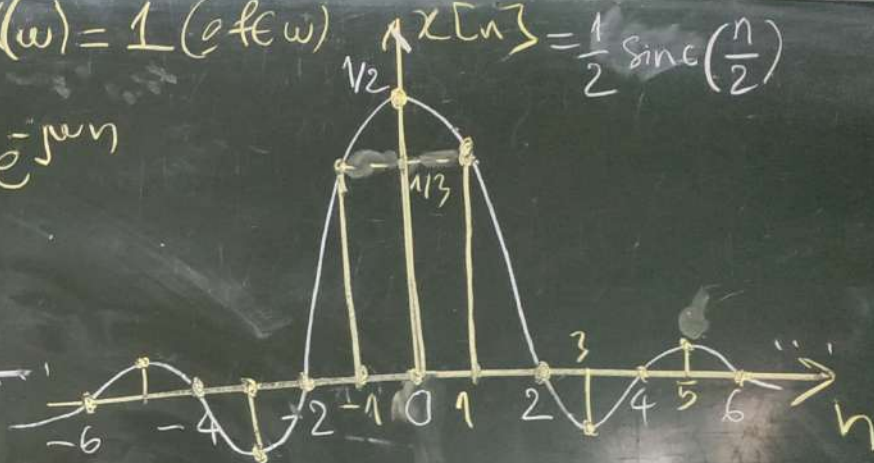
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e) C2: Hệ quả: (Mặc $x[n] = \delta[n]$) $\xrightarrow{\text{DTFT}} X(\omega) = 1 \quad (\forall \omega)$

(vi) $X(\omega) = \sum_{n=-\infty}^{+\infty} x[n] \cdot e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} \delta[n] \cdot e^{-j\omega n}$
 $= \delta[0] \cdot e^{-j\omega \cdot 0} = 1$

$x[n] = \frac{1}{2} \text{sinc}\left(\frac{n}{2}\right)$



AD hệ quả trên và TC dịch t/g ω DTFT:

$$x_1[n] = \frac{1}{2} x[n] = \frac{1}{2} \delta[n] \xrightarrow{\text{DTFT}} X_1(\omega) = \frac{1}{2} X(\omega) = \frac{1}{2}$$

$$x_2[n] = \frac{1}{4} x[n - (-2)] = \frac{1}{4} \delta[n+2] \xrightarrow{\text{DTFT}} X_2(\omega) = \frac{1}{4} e^{j2\omega} = \frac{1}{4} \cdot e^{-j\omega(-2)} \cdot 1 = \frac{1}{4} e^{-j\omega(-2)} \cdot X(\omega)$$

$$x_3[n] = \frac{1}{4} x[n-2] = \frac{1}{4} \delta[n-2] \xrightarrow{\text{DTFT}} X_3(\omega) = \frac{1}{4} e^{-j\omega \cdot 2} \cdot 1 = \frac{1}{4} e^{-j2\omega} \cdot X(\omega)$$

$$\begin{aligned} \rightarrow x[n] &= x_1[n] + x_2[n] + x_3[n] \\ &= \frac{1}{2} \delta[n] + \frac{1}{4} \delta[n+2] + \frac{1}{4} \delta[n-2] \end{aligned}$$



TC dịch t/g u/v DTFT: (Textbook/Slide)

$$\text{Nếu } x[n] \xrightarrow{\text{DTFT}} X(e^{j\omega}) \equiv X(\omega)$$

$$\text{thì } x[n-n_0] \xrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega}) \quad \forall n_0 \in \mathbb{Z}$$

$$\underline{1:} X(\omega) = \cos^2 \omega = \frac{1}{2} (1 + \cos 2\omega)$$

$$= \frac{1}{2} \left(1 + \frac{1}{2} (e^{j2\omega} + e^{-j2\omega}) \right)$$